Horticulture Project and Seminar

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Horticulture Project and Seminar (A14579)

Short Title: Hort Project.
Faculty Science and Computing
Department Science
Cluster Placement and Projects
Author CDALY
Credits 5 **Level:** Intermediate

Description of Module / Aims

This module provides the opportunity to collect, analyse and present horticulture-related data obtained from a group project. Projects and groups will be assigned to students. Projects may involve growth experiments, surveys of societal or consumer attitudes or inclinations, or may be design oriented. Projects will be implemented by the group and written up individually. Groups will then work together at the end of the semester to present their findings in a poster seminar.

Programmes

HORT-0044	BSc in Horticulture (WD_SHORT_D)	3 / 6 / M
HORT-0044	BSc in Horticulture (WD_SHORT_DX)	3 / 6 / M

Indicative Content

- The scientific method(s): generating and using null and alternative hypotheses; examining what is a researchable question
- Working effectively as part of a team: what to expect from your peers, from your supervisor, and methods to overcome motivational challenges within a team
- Structuring a research project: deciding number of replicates, random block design generation, positive and negative controls, working titles, project outline, and timing
- Collecting and interpreting data: designing and administering questionnaires, planning and conducting interviews, using internet-based facilities to share group data, and pitfalls to avoid when interpreting data
- Presenting results: graphical and tabular presentation of data, and explanation of data quality using standard deviation and standard error
- Designing scientific posters: generating impactful, clear and attractive poster presentations
- Presentation skills: examining the effective use of narrative-based presentation of scientific data; methods of overcoming performance anxiety and building resilience
- Writing reports: getting started, structuring, and revising written reports, guarding against plagiarism and evaluating the written report

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Formulate a horticulture-related hypothesis using the scientific method.
2. Set up a horticultural experiment and collect data as part of a team.
3. Prepare a report outlining methods, results and significance of the outcomes of a horticulture-related study.
4. Design, and present a poster presentation as part of a team on a body of research.
5. Appraise the methods used and data collected during the course of a horticulture-related study.
6. Establish the limitations and opportunities for improvement within a horticulture-related study.

Learning and Teaching Methods

- Lectures.
- Group and individual workshops.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Project	60%	1,3,5,6
Participation	20%	2,4,
Presentation	20%	4

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to work effectively as a researcher. Functions poorly as a member of a team.

40%-49%: Demonstrates a limited knowledge of scientific method(s) and experiment design and implementation. Shows some technical ability/skill in carrying out and reporting on practical tasks. Limited function as a member of a team.

50%-59%: Demonstrates satisfactory general knowledge of scientific method(s) and experiment design and implementation. Logically and competently carries out practical tasks. Works well as a member of the team and is often one of the first to show initiative in a group.

60%-69%: Demonstrates sound knowledge of experiment design and implementation of experiments according to scientific methods. Practicals and reports completed to high standard. Works effectively as a team member and shows continued initiative within a working group.

70%-100%: Demonstrates excellent technical ability, awareness, and aptitude for working as a researcher. Delivers insightful scientific writing and presents data with impact. Develops high competency in data analysis and shows leadership qualities during group work.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Workshop	12	
Independent Learning	111	

Essential Material

- Bell, J. and S. Waters. _Doing Your Research Project: A Guide for First-Time Researchers in Education, Health and Social Science_. 6th ed. Berkshire, UK: Open University Press, 2005.

Supplementary Material

- Jones, A., R. Reed and J. Weyers. _Practical Skills in Biology_. Oxford, UK. : Pearson, 2012.
- Pechenik, J.A. _A Short Guide to Writing About Biology_. 9th ed. London, Uk: Pearson, 2016.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Computer Lab